MTH 201 Chapter 2: Integration Techniques

2.6 Worksheet

1. Approximate the integral $\int_0^3 \sqrt{x+1}\,dx$ using…

a) a midpoint approximation with $n=6$. Round your answer to two decimals.

$$\Delta x = \frac{3-0}{6} = \frac{1}{2}$$

$$\int_0^3 \sqrt{x+1}dx \approx \frac{1}{2}\left[f(0.25)+f(0.75)+f(1.25)+f(1.75)+f(2.25)+f(2.75)\right]$$

$$= \frac{1}{2}\left[\sqrt{1.25}+\sqrt{1.75}+\sqrt{2.25}+\sqrt{2.75}+\sqrt{3.25}+\sqrt{3.75}\right]$$

$$\approx 4.67$$

b) a trapezoidal approximation with $n=3$. Round your answer to two decimals.

$$\Delta x = \frac{3-0}{3} = 1$$

$$\int_0^3 \sqrt{x+1}dx \approx \frac{1}{2}\left[f(0)+2f(1)+2f(2)+f(3)\right]$$

$$= \frac{1}{2}\left[\sqrt{1}+2\sqrt{2}+2\sqrt{3}+2\sqrt{4}\right]$$

$$\approx 5.65$$

c) Simpson's Rule using $n=6$. Round your answer to two decimals.

$$\Delta x = \frac{3-0}{6} = \frac{1}{2}$$

$$\int_0^3 \sqrt{x+1}dx \approx \frac{1/2}{3}\left[f(0)+4f(0.5)+2f(1)+4f(1.5)+2f(2)+4f(2.5)+f(3)\right]$$

$$= \frac{1}{6}\left[\sqrt{1}+4\sqrt{1.5}+2\sqrt{2}+4\sqrt{2.5}+2\sqrt{3}+4\sqrt{3.5}+2\sqrt{4}\right]$$

$$\approx 5.00$$

2. Consider the integral $\int_2^4 \frac{1}{x^3}\,dx$.

a) Approximate the value of this integral using a midpoint approximation with 4 rectangles. Round your answer to four decimals.

$$\Delta x = \frac{4-2}{4} = \frac{1}{2}$$

$$\int_2^4 \frac{1}{x^3}\,dx \approx \frac{1}{2}\left[f(2.25)+f(2.75)+f(3.25)+f(3.75)\right] \approx 0.0920$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $|E_M| \le \frac{K(b-a)^3}{24n^2}$ where K is some value such that $|f''(x)| \le K$ on $[a,b]$.

We calculate that $f''(x) = 12x^{-5}$ and so we need a value of K such that $\left|\frac{12}{x^5}\right| \le K$ on $[2,4]$. We let $K = \frac{3}{8}$.

Then we have that $|E_M| \le \frac{K(b-a)^3}{24n^2} = \frac{\frac{3}{8}(4-2)^3}{24(4)^2} = \frac{3}{384} = \frac{1}{128} \approx 0.01$.

This means that the approximation we made in part (a) is, at worst, roughly 0.01 away from correct value of the integral $\int_2^4 \frac{1}{x^3}\,dx$.

ii. How large must n be to ensure an error of no more than $\frac{1}{80,000} = 0.0000125$?

So, we want to have a value of n that makes $|E_M| \le \frac{K(b-a)^3}{24n^2} < \frac{1}{80000}$

We still use the values of a, b, and K from part (i) to set up the inequality

$$\frac{\frac{3}{8}(4-2)^3}{24n^2} < \frac{1}{80000} \;\rightarrow\; 24n^2 > 240000 \;\rightarrow\; n^2 > 10000 \;\rightarrow\; n > \sqrt{10000} = 100$$

Therefore, n must be at least 101.

b) Approximate the value of this integral using a trapezoidal approximation with 4 trapezoids. Round your answer to four decimals.

$$\Delta x = \frac{4-2}{4} = \frac{1}{2}$$

$$\int_2^4 \frac{1}{x^3}\,dx \approx \frac{\left(\frac{1}{2}\right)}{2}\left[f(2)+2f(2.5)+2f(3)+2f(3.5)+f(4)\right] \approx 0.0973$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $|E_M| \le \frac{K(b-a)^3}{24n^2}$ where K is some value such that $|f''(x)| \le K$ on $[a,b]$.

We calculate that $f''(x) = 12x^{-5}$ and so we need a value of K such that $\left|\frac{12}{x^5}\right| \le K$ on $[2,4]$. We let $K = \frac{3}{8}$.

Then we have that $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{\frac{3}{8}(4-2)^3}{12(4)^2} = \frac{3}{192} = \frac{1}{64} \approx 0.02$.

This means that the approximation we made in part (a) is, at worst, roughly 0.02 away from correct value of the integral $\int_2^4 \frac{1}{x^3}\,dx$.

ii. How large must n be to ensure an error of no more than $\frac{1}{80000} = 0.0000125$?

So, we want to have a value of n that makes $|E_T| \le \frac{K(b-a)^3}{12n^2} < \frac{1}{80000}$

We still use the values of a, b, and K from part (i) to set up the inequality

$$\frac{\frac{3}{8}(4-2)^3}{12n^2} < \frac{1}{80000} \;\rightarrow\; 12n^2 > 240000 \;\rightarrow\; n^2 > 20000 \;\rightarrow\; n > \sqrt{20000} \approx 141.42$$

Therefore, n must be at least 142.

c) Approximate the value of this integral using Simpson's Rule with $n = 4$. Round your answer to two decimals.

$$\Delta x = \frac{4-2}{4} = \frac{1}{2}$$

$$\int_2^4 \frac{1}{x^3}\,dx \approx \frac{1}{3}\left(\frac{2}{4}\right)\left[f(2)+4f(2.5)+2f(3)+4f(3.5)+f(4)\right] \approx 0.0940$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $\left|E_M\right| \le \frac{K(b-a)^3}{24n^2}$ where K is some value such that $\left|f^{(4)}(x)\right| \le K$ on $[a,b]$.

We calculate that $f^{(4)}(x) = 360x^{-7}$ and so we need a value of K such that $\left|\frac{360}{x^7}\right| \le K$ on $[2,4]$.

We let $K = \frac{45}{16}$.

Then we have that $\left|E_S\right| \le \frac{K(b-a)^5}{180n^4} = \frac{\frac{45}{16}(4-2)^5}{180(4)^4} = \frac{90}{2880} = \frac{9}{288} = \frac{1}{32} \approx 0.03$.

This means that the approximation we made in part (a) is, at worst, roughly 0.03 away from correct value of the integral $\int_2^4 \frac{1}{x^3}\,dx$.

ii. How large must *n* be to ensure an error of no more than $\frac{1}{80000} = 0.0000125$?

So, we want to have a value of *n* that makes $\left|E_S\right| \le \frac{K(b-a)^5}{180n^4} < \frac{1}{80000}$

We still use the values of *a*, *b*, and *K* from part (i) to set up the inequality

$$\frac{\frac{45}{16}(4-2)^5}{180n^4} < \frac{1}{80000} \quad \rightarrow \quad 180n^4 > 240000 \quad \rightarrow \quad n^4 > 400000 \quad \rightarrow \quad n > \sqrt[4]{400000} \approx 25.15$$

Therefore, *n* must be at least 26 (since n must always be even with Simpson's Rule).

3. Consider the integral $\int_0^{\pi/2} \cos(x)\,dx$.

a) Approximate the value of this integral using a midpoint approximation with 3 rectangles. Round your answer to two decimals.

$$\Delta x = \frac{\frac{\pi}{2}-0}{3} = \frac{\pi}{6}$$

$$\begin{aligned}\int_0^{\pi/2} \cos(x)\,dx &\approx \frac{\pi}{6}\left[f\left(\frac{\pi}{12}\right)+f\left(\frac{3\pi}{12}\right)+f\left(\frac{5\pi}{12}\right)\right] \\ &= \frac{\pi}{6}\left[\cos\left(\frac{\pi}{12}\right)+\cos\left(\frac{3\pi}{12}\right)+\cos\left(\frac{5\pi}{12}\right)\right] \\ &\approx 1.01\end{aligned}$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $|E_M| \le \frac{K(b-a)^3}{24n^2}$ where K is some value such that $|f''(x)| \le K$ on $[a,b]$. We calculate that $f''(x) = -\cos(x)$ and so we need a value of K such that $|\cos(x)| \le K$ on $\left[0, \frac{\pi}{2}\right]$. We let $K = 1$.

Then we have that $|E_M| \le \frac{K(b-a)^3}{24n^2} = \frac{1\left(\frac{\pi}{2}-0\right)^3}{24(3)^2} = \frac{\pi^3/8}{216} = \frac{\pi^3}{1728} \approx 0.02$.

This means that the approximation we made in part (a) is, at worst, roughly 0.02 away from correct value of the integral $\int_0^{\pi/2} \cos(x)\,dx$.

ii. How large must n be to ensure an error of no more than 0.0001?

So, we want to have a value of n that makes $|E_M| \le \frac{K(b-a)^3}{24n^2} < 0.0001$

We still use the values of a, b, and K from part (i) to set up the inequality

$$\frac{1\left(\frac{\pi}{2}-0\right)^3}{24n^2} < 0.0001 \;\rightarrow\; 24n^2 > \frac{10000\pi^3}{8} \;\rightarrow\; n^2 > \frac{10000\pi^3}{8\cdot 24} \;\rightarrow\; n > \sqrt{\frac{10000\pi^3}{8\cdot 24}} \approx 40.19$$

Therefore, n must be at least 41.

b) Approximate the value of this integral using a trapezoidal approximation with 3 trapezoids. Round your answer to two decimals.

$$\Delta x = \frac{\frac{\pi}{2} - 0}{3} = \frac{\pi}{6}$$

$$\int_0^{\pi/2} \cos(x)\,dx \approx \frac{\pi/6}{2}\left[f(0) + 2f\left(\frac{\pi}{6}\right) + 2f\left(\frac{2\pi}{6}\right) + 2f\left(\frac{\pi}{2}\right)\right]$$

$$= \frac{\pi}{12}\left[\cos(0) + 2\cos\left(\frac{\pi}{6}\right) + 2\cos\left(\frac{\pi}{3}\right) + 2\cos\left(\frac{\pi}{2}\right)\right]$$

$$= \frac{\pi}{12}\left[1 + \sqrt{3} + 1 + 0 \right]$$

$$\approx 0.98$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $|E_T| \le \frac{K(b-a)^3}{12n^2}$ where K is some value such that $|f''(x)| \le K$ on $[a,b]$. We calculate that $f''(x) = -\cos(x)$ and so we need a value of K such that $|\cos(x)| \le K$ on $\left[0, \frac{\pi}{2}\right]$. We let $K = 1$.

Then we have that $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{1\left(\frac{\pi}{2} - 0\right)^3}{12(3)^2} = \frac{\pi^3/8}{108} = \frac{\pi^3}{864} \approx 0.04$.

This means that the approximation we made in part (a) is, at worst, roughly 0.04 away from correct value of the integral $\int_0^{\pi/2} \cos(x)\,dx$.

ii. How large must n be to ensure an error of no more than 0.0005?

So, we want to have a value of n that makes $|E_T| \le \frac{K(b-a)^3}{12n^2} < 0.0005$

We still use the values of a, b, and K from part (i) to set up the inequality

$$\frac{1\left(\frac{\pi}{2} - 0\right)^3}{12n^2} < 0.0005 \;\rightarrow\; 12n^2 > \frac{2000\pi^3}{8} \;\rightarrow\; n^2 > \frac{2000\pi^3}{8\cdot 12} \;\rightarrow\; n > \sqrt{\frac{2000\pi^3}{8\cdot 12}} \approx 25.42$$

Therefore, n must be at least 26.

c) Approximate the value of this integral using Simpson's Rule with $n = 4$. Round your answer to two decimals.

$$\Delta x = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}$$

$$\begin{aligned}\int_0^{\pi/2} \cos(x)\,dx &\approx \frac{\pi/8}{3}\left[f(0) + 4f\left(\frac{\pi}{8}\right) + 2f\left(\frac{2\pi}{8}\right) + 4f\left(\frac{3\pi}{8}\right) + f\left(\frac{\pi}{2}\right)\right] \\ &= \frac{\pi}{24}\left[\cos(0) + 4\cos\left(\frac{\pi}{8}\right) + 2\cos\left(\frac{\pi}{4}\right) + 4\cos\left(\frac{3\pi}{8}\right) + \cos\left(\frac{\pi}{2}\right)\right] \\ &\approx 1.00\end{aligned}$$

i. Determine an upper bound on the error of this approximation using the error estimation formula discussed in the notes. Round this upper bound to 2 places and interpret this value.

The error formula is $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$ where K is some value such that $\left|f^{(4)}(x)\right| \le K$ on $[a,b]$.

We calculate that $f^{(4)}(x) = \cos(x)$ and so we need a value of K such that $|\cos(x)| \le K$ on $\left[0, \frac{\pi}{2}\right]$. We let $K = 1$.

Then we have that $|E_S| \le \dfrac{K(b-a)^5}{180n^4} = \dfrac{1\left(\frac{\pi}{2} - 0\right)^5}{180(4)^4} = \dfrac{\pi^5/32}{2880} = \dfrac{\pi^5}{92160} \approx 0.00$.

This means that the approximation we made in part (a) is, at worst, roughly 0.00 away from correct value of the integral $\int_0^{\pi/2} \cos(x)\,dx$.

ii. How large must n be to ensure an error of no more than $\dfrac{1}{100}$?

So, we want to have a value of n that makes $|E_S| \le \dfrac{K(b-a)^5}{180n^4} < 0.01$

We still use the values of a, b, and K from part (i) to set up the inequality

$$\frac{1\left(\frac{\pi}{2} - 0\right)^5}{180n^4} < 0.01 \;\rightarrow\; 180n^4 > \frac{100\pi^5}{32} \;\rightarrow\; n^4 > \frac{100\pi^5}{32\cdot 180} \;\rightarrow\; n > \sqrt[4]{\frac{100\pi^5}{32\cdot 180}} \approx 1.52$$

Therefore, n must be at least 2.